Hyperparameter Tuning

Finding the Perfect Recipe for Your ML Model

Grid Search | Random Search | MLflow Nested Runs | Model Registry

MLOps Lab 6 · QuickFoods Case Study

What Are Hyperparameters?

🛵 The QuickFoods Analogy

Goal: deliver every order in under 10 minutes

Past orders (fixed data) 📦
History of deliveries — can't change what already happened

Order details (features) 📋
distance, items, traffic, peak hour — inputs to the model

Drivers on shift 👥
n_estimators — more drivers = more coverage, costs more compute

Max stops per driver 🗺️
max_depth — too many stops and the driver gets lost in complexity

Rerouting aggression 🚦
learning_rate — how fast the model reacts to new traffic patterns

The Definition

Hyperparameters are settings you choose BEFORE training starts. The model cannot learn them on its own — you must set them.

Examples in this Lab

`n_estimators`	How many trees in the forest?
`max_depth`	How tall can each tree grow?
`learning_rate`	How big are the learning steps?
`min_samples_split`	Min data to split a branch?

Grid Search — Try Every Single Combination

🍦 Analogy: The Ice Cream Shop

Imagine you want the BEST sundae. You have 3 flavours × 3 toppings × 2 sauces = 18 combos. Grid Search tastes ALL 18. Slow, but guaranteed to find the best one in the list!

QuickFoods RF Grid (3 × 3 × 2 = 18 trials)

`n_estimators`	50	100	200
`max_depth`	5	10	None
`min_samples_split`	2	5	

How It Works — Step by Step

1. Define a grid of values for each parameter
2. Generate every possible combination
3. Train a model for each combination
4. Measure MAE on the test set
5. Pick the combo with the lowest MAE

✓ Exhaustive

✗ Slow if grid is large

Random Search & MLflow Nested Runs

🗺️ Random Search

Analogy: Treasure Hunt on a Map

Imagine the map has 18 X marks (like Grid Search). You only have time to dig 6 holes — so you pick 6 spots at RANDOM. You might not find the ABSOLUTE best treasure, but you'll likely find something very good, much faster!

Side-by-Side Comparison

Method	Trials	Guarantee
Grid Search	All 18 combos	Guaranteed best in grid
Random Search	Only 6 trials	Fast, usually very good

Use Random Search when the parameter space is large and you have limited compute budget.

📁 MLflow Nested Runs

Analogy: Filing Cabinet

The cabinet drawer = Parent Run (the whole sweep). Each folder inside = one Child Run (one trial). Every child keeps its own params, metrics, and model — nothing gets mixed up!

```
🗂️ HyperparamSweep-QuickFoods  (Parent)
    ├ RF | n_estimators=50   MAE=4.81
    ├ RF | n_estimators=100  MAE=4.32
    ├ GB | lr=0.2  MAE=4.35
    └ GB | lr=0.1  MAE=4.10  ⭐ BEST
```

Every trial is saved forever — reproducible, auditable, comparable.

Selecting & Registering the Winner 🏆

1
Search all child runs

Filter runs with tag sweep_child = true. Sort by MAE ascending.

2
Identify the best

candidates[0] is the run with the lowest MAE across all trials.

3
Register in MLflow

mlflow.register_model() creates a versioned entry in the Model Registry.

4
Promote to Production

Version gets a lifecycle stage: Staging → Production. Serves via FastAPI in Lab 7.

🍱 *Think of it like a cooking contest: every chef's dish (trial) is tasted and scored (MAE). The winner's recipe goes on the restaurant menu (Model Registry) and customers order it every day (production API).*